VECTORS · TWO PRODUCTS

Dot against cross

Two ways of multiplying vectors that disagree on almost every attribute. Where one is zero the other is largest, and between them they measure the two things a pair of vectors has: how much they agree, and how much they span.

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PRODUCTS

TYPE	CONDITION	OUTPUT	DEFINED IN	SWAPPING THE ARGUMENTS	ZERO WHEN	GEOMETRIC READIN
MEASURES AGREEMENT						
 **Dot product**	$\mathbf{a} \cdot \mathbf{b} = \lVert\mathbf{a}\rVert\lVert\mathbf{b}\rVert \cos\theta$	a scalar	any $\mathbb{R}^n$	unchanged — commutative	perpendicular, $\theta = \pi/2$	projection of one onto the other
MEASURES SPREAD						
 **Cross product**	$\lVert\mathbf{a} \times \mathbf{b}\rVert = \lVert\mathbf{a}\rVert\lVert\mathbf{b}\rVert \sin\theta$	a vector	only $\mathbb{R}^3$	sign reverses — anti-commutative	parallel, $\theta = 0$ or π	area of the parallelogram spanned

The opposition is systematic rather than coincidental. One extracts $\cos\theta$ and one extracts $\sin\theta$, and every difference below follows: zero at opposite angles, scalar against vector, commutative against sign-reversing. Given both products, θ is determined completely — neither alone suffices.